

Vapor, Barriers and Wet Insulation

Permeability, which side the barrier goes on, and how to read a roof that only leaks in winter.

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Permeability is the property of a material that allows the passage of:
Water vapor.

Permeability is specifically about vapor, which is exactly why it is the property that defines a vapor retarder. Low perm blocks vapor; high perm lets an assembly dry out.

Vapor retarder	Belongs on the warm-in-winter side of the insulation — the interior face in a cold climate. Vapor travels from warm to cold, so the barrier goes where the vapor starts.
What it prevents	Moisture building up inside the exterior wall. Warm indoor air carrying moisture reaches a cold surface inside the wall and condenses there.
Not a vapor barrier	Wood. Polyethylene sheet, foil facing and kraft paper facing are; house wrap and wood are not.

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The vapor barrier prevents:
Moisture from building up inside the exterior wall.

A roof that leaks only in winter

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If a metal roof seems to have many leaks but no specific leakage area, and it only leaks in the winter, you should consider:
Condensation on the underside of the metal.

Two clues point away from a leak and toward condensation: there is no single source, and it is seasonal. Metal goes cold fast, indoor moisture reaches the cold underside and condenses along the whole surface. Water appearing everywhere at once, only in cold weather, is not a hole in the roof.

Why attic insulation gets wet

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Attic insulation may be wet as a result of any of the following EXCEPT:
Insulation that is too close to chimneys.

Insulation packed against a chimney is a fire-clearance problem, not a moisture one. The genuine wetting causes are roof leaks, condensation on cold surfaces, and bath or dryer exhaust dumped into the attic. On an EXCEPT question, find the one that belongs to a different category of fault altogether.

Say it back

Cover the page. Answer each one out loud.

1. What does permeability measure?
2. Which side of the insulation does the vapor retarder go on, and why?
3. What does the vapor barrier prevent?
4. Name a material that is NOT used as a vapor barrier.
5. A metal roof leaks everywhere, but only in winter. What do you consider?
6. Name the real causes of wet attic insulation, and the one that is not.